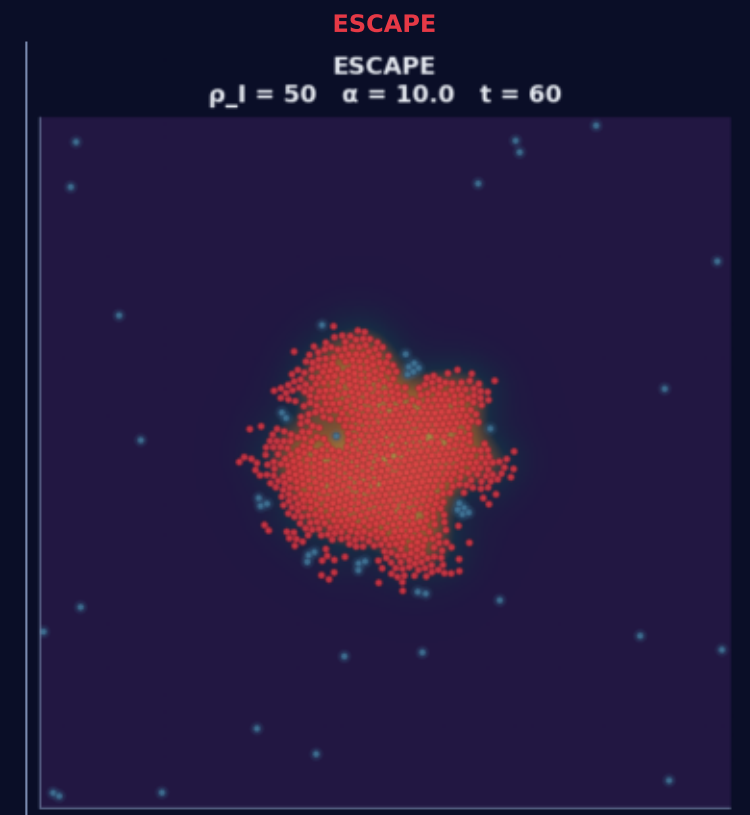
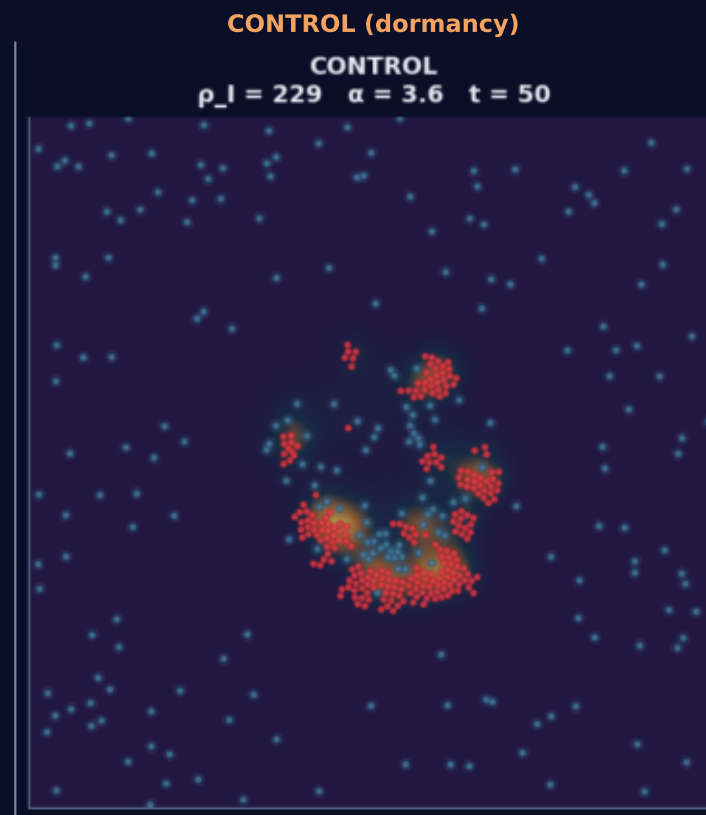
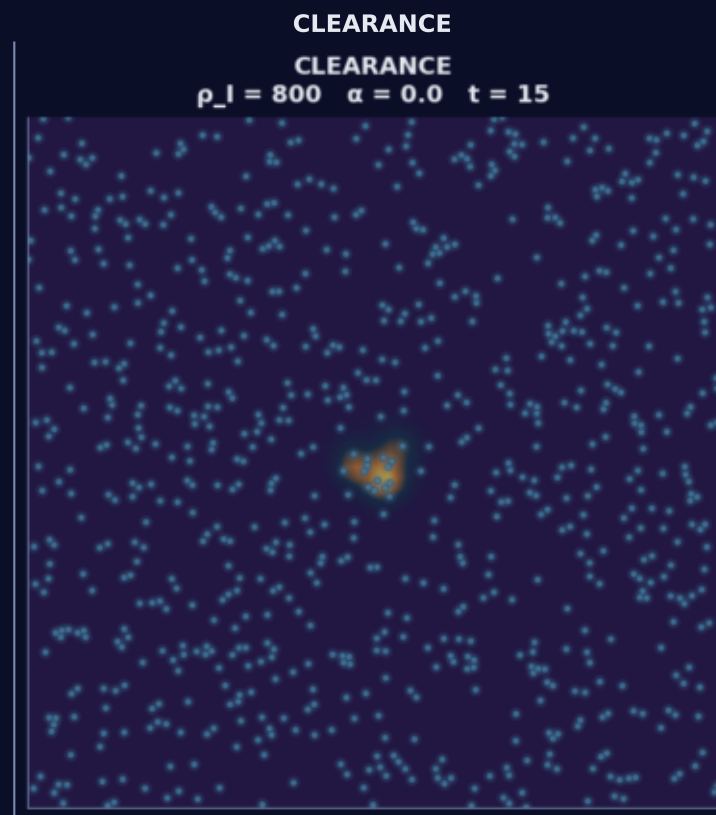


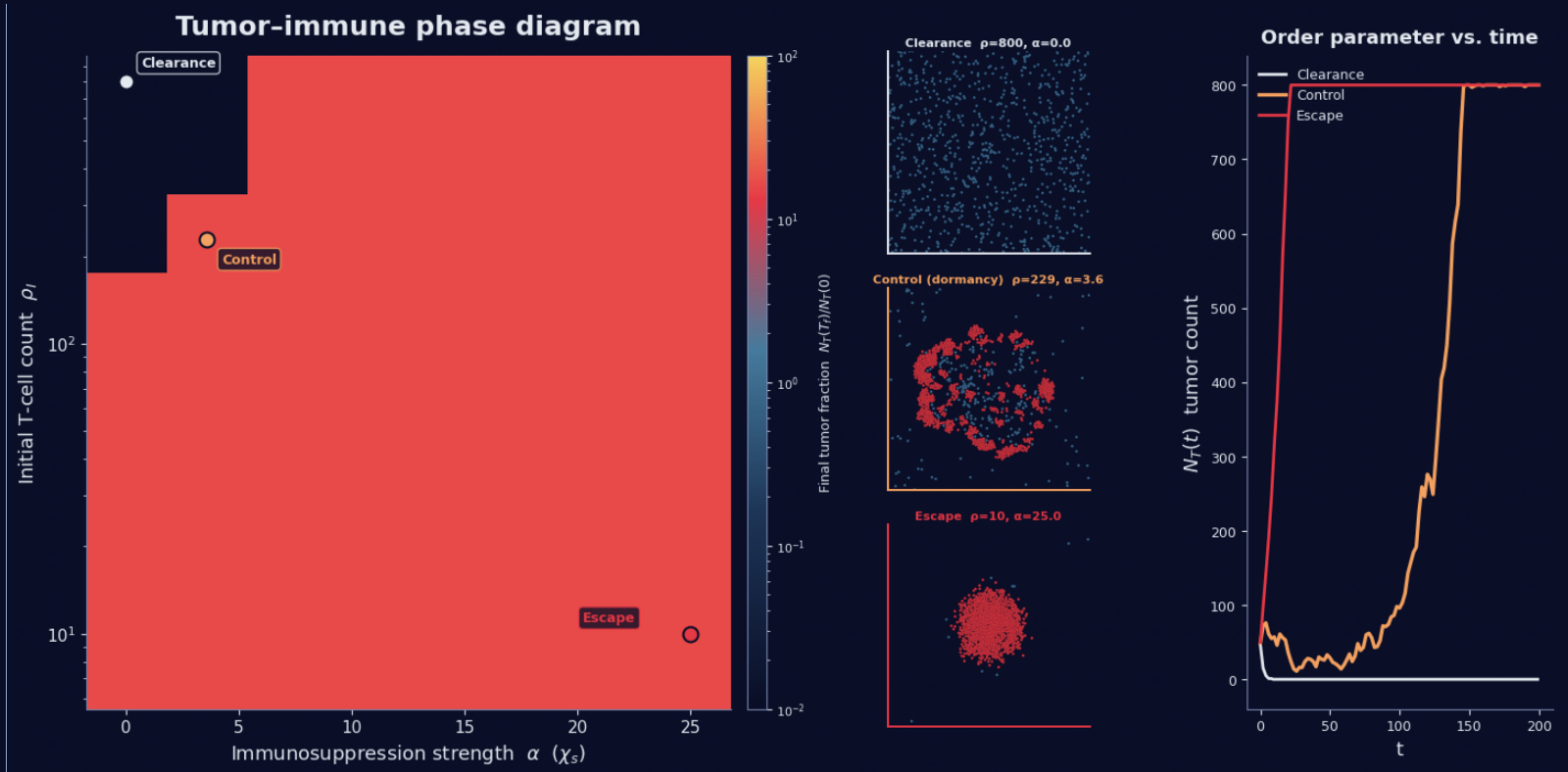
Tumor-Immune Active Matter: three phases from minimal rules

Slow proliferating tumor cells secrete an attractant (recruits T cells) and a short-range suppressant (repels them). The balance produces three regimes matching the clinical 'hot / excluded / cold' classification of solid tumors.



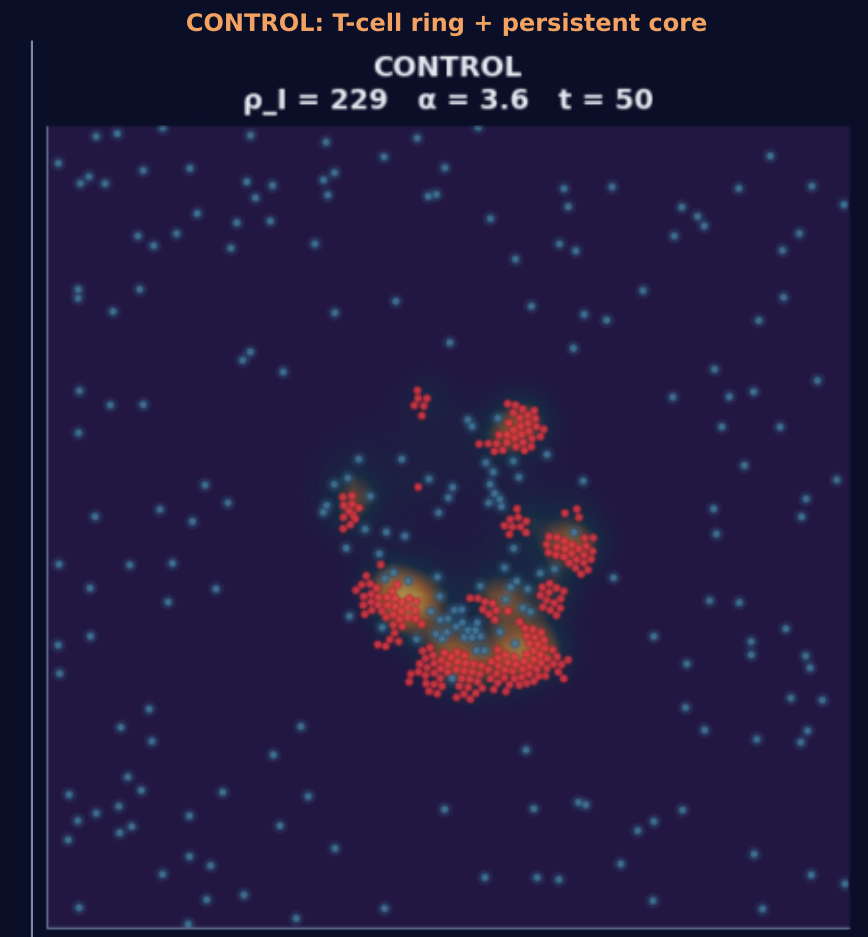
Phase diagram in (ρ_I , α) space

Sweep over initial T-cell count (ρ_I) and immunosuppression strength ($\alpha = \chi_s$).
Each cell: geometric mean of 3 seeds. Final tumor fraction on log color scale.



Mechanism: an activator-inhibitor race

- Tumor cells secrete BOTH long-range attractant c_a ($D_a = 5$) and short-range suppressant c_s ($D_s = 0.5$). T cells feel $+\chi_a \nabla c_a - \alpha \nabla c_s$.
- Three competing rates set the outcome at any (ρ_I, α) :
 - proliferation rate $p_{div} \cdot N_T$
 - T-cell recruitment $\chi_a \times \text{tumor } c_a \text{ flux}$ (grows with N_T)
 - T-cell exclusion $\alpha \times \text{tumor } c_s \text{ flux}$ (also grows with N_T)
- In CLEARANCE: recruitment dominates exclusion, T cells reach tumor surface fast enough that kill rate $>$ birth rate; tumor extinct in ~ 10 time units.
- In CONTROL (dormancy): the two field couplings are nearly balanced. Tumor is pinned 50–100 cells; small fluctuations either clear it or trigger escape.
- In ESCAPE: suppressant repels T cells faster than attractant pulls them in. Tumor mass grows unimpeded; T cells form a frustrated ring outside.



Biological mapping

Simulation phase	Clinical phenotype	Histology signature
Clearance (low α)	'Hot' tumor	Infiltrating CD8+ T cells
Control / dormancy	Immune-excluded tumor	T cells in a ring around the core
Escape (high α)	'Cold' tumor	Few intratumoral T cells; immunosuppressive stroma

Mechanistic interpretation of immunotherapy:

Anti-PD-1 / anti-PD-L1 / anti-CTLA-4 antibodies reduce effective immunosuppression strength α . In our model: lowering α moves a tumor from the escape phase into the control or clearance phase. The treatment experiment (next slide) demonstrates this rescue.

Treatment experiment: α -knockdown rescues an escape tumor

Identical run, identical seed. At $t = 20$, suppressant chemotaxis coupling α is set from $10 \rightarrow 0$ (full checkpoint inhibition). Trajectory pivots immediately.

Simulated checkpoint inhibitor rescues an escape-phase tumor

